SCHOOLOFCOMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENTOFCOMPUTER SCIENCE ENGINEERING	
ProgramName: B. Tech		**AssignmentType: Lab**	**AcademicYear:** 2025-2026
CourseCoordinatorName		Venkataramana Veeramsetty	
Instructor(s)Name		Dr. V. Venkataramana (Co-ordinator) Dr. T. Sampath Kumar Dr. Pramoda Patro Dr. Brij Kishor Tiwari Dr.J.Ravichander Dr. Mohammand Ali Shaik Dr. Anirodh Kumar Mr. S.Naresh Kumar Dr. RAJESH VELPULA Mr. Kundhan Kumar Ms. Ch.Rajitha Mr. M Prakash Mr. B.Raju Intern 1 (Dharma teja) Intern 2 (Sai Prasad) Intern 3 (Sowmya) NS_2 (Mounika)	
CourseCode	24CS002PC215	**CourseTitle**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
DateandDay of Assignment	Week4 - Thursday	**Time(s)**	
Duration	2 Hours	**Applicableto Batches**	
AssignmentNumber: **7.4**(Presentassignmentnumber)/**24**(Totalnumberofassignments)			

Q.No.	Question	*ExpectedTime to complete*
1	Lab 7: Error Debugging with AI – Systematic Approaches to Finding and Fixing Bugs Lab Objectives: • To identify and correct syntax, logic, and runtime errors in Python programs using AI tools.	Week4 - Thursday

- To understand common programming bugs and AI-assisted debugging suggestions.
- To evaluate how AI explains, detects, and fixes different types of coding errors.
- To build confidence in using AI to perform structured debugging practices.

Lab Outcomes (LOs):
After completing this lab, students will be able to:

- Use AI tools to detect and correct syntax, logic, and runtime errors.
- Interpret AI-suggested bug fixes and explanations.
- Apply systematic debugging strategies supported by AI-generated insights.
- Refactor buggy code using responsible and reliable programming patterns.

Task Description #1:
• Introduce a buggy Python function that calculates the factorial of a number using recursion. Use Copilot or Cursor AI to detect and fix the logical or syntax errors.

```
def factr(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return n * factr(n - 2)

print(factr("5"))
```

Expected Outcome #1:
•Copilot or Cursor AI correctly identifies missing base condition or incorrect recursive call and suggests a functional factorial implementation.

Task Description #2:
•Provide a list sorting function that fails due to a type error (e.g., sorting list with mixed integers and strings). Prompt AI to detect the issue and fix the code for consistent sorting**.**

```
def sort_list(data):
    return sorted(data)

items = [3, "apple", 1, "banana", 2]
print(sort_list(items))
```

Expected Outcome #2:
•AI detects the type inconsistency and either filters or converts list elements, ensuring successful sorting without a crash.

Task Description #3:
• Write a Python snippet for file handling that opens a file but forgets to close it. Ask Copilot or Cursor AI to improve it using the best practice (e.g., with open() block).

Code1

```
with open("example.txt", "w") as f:
    f.write("Hello, world!")
```

Code2

```
f1 = open("data1.txt", "w")
f2 = open("data2.txt", "w")

f1.write("First file content\n")
f2.write("Second file content\n")

print("Files written successfully")
```

Code3

```
data = open("input.txt", "r").readlines()
output = open("output.txt", "w")

for line in data:
    output.write(line.upper())

print("Processing done")
```

Code4:

```
f = open("numbers.txt", "r")
nums = f.readlines()

squares = []
for n in nums:
    n = n.strip()
    if n.isdigit():
        squares.append(int(n) * int(n))

f2 = open("squares.txt", "w")
for sq in squares:
    f2.write(str(sq) + "\n")

print("Squares written")
```

Expected Outcome #3:
• AI refactors the code to use a context manager, preventing resource leakage and runtime warnings.

Task Description #4:
• Provide a piece of code with a ZeroDivisionError inside a loop. Ask AI to add error handling using try-except and continue execution safely.

```
def compute_ratios(values):
    results = []
    for i in range(len(values)):
        for j in range(i, len(values)):
            ratio = values[i] / (values[j] - values[i])
            results.append((i, j, ratio))
    return results

nums = [5, 10, 15, 20, 25]
print(compute_ratios(nums))
```

Expected Outcome #4:
• Copilot adds a try-except block around the risky operation, preventing crashes and printing a meaningful error message.

Task Description #5:
• Include a buggy class definition with incorrect __init__ parameters or attribute references. Ask AI to analyze and correct the constructor and attribute usage.

```
class StudentRecord:
    def __init__(self, name, id, courses=[]):
        self.studentName = names
        self.student_id = id
        self.courses = courseList

    def add_course(self, course):
        self.courses.append(course)

    def get_summary(self):
        return f"Student: {self.studentName}, ID: {self.student_id}, Courses: {', '.join(self.courses)}"

class Department:
    def __init__(self, deptName, students=None):
        self.dept_name = deptName
        self.students = students

    def enroll_student(self, student):
        self.students.append(student)

    def department_summary(self):
        return f"Department: {self.dept_name}, Total Students: {len(self.student)}"


s1 = StudentRecord("Alice", 101, ["Math", "Science"])
d1 = Department("Computer Science")
d1.enroll_student(s1)
print(s1.get_summary())
print(d1.department_summary())
```

Expected Outcome #5:
• Copilot identifies mismatched parameters or missing self references and rewrites the class with accurate initialization and usage.

Task Description #1:

• Introduce a buggy Python function that calculates the factorial of a number using recursion. Use Copilot or Cursor AI to detect and fix the logical or syntax errors.

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def factr(n):
    if n == 0:
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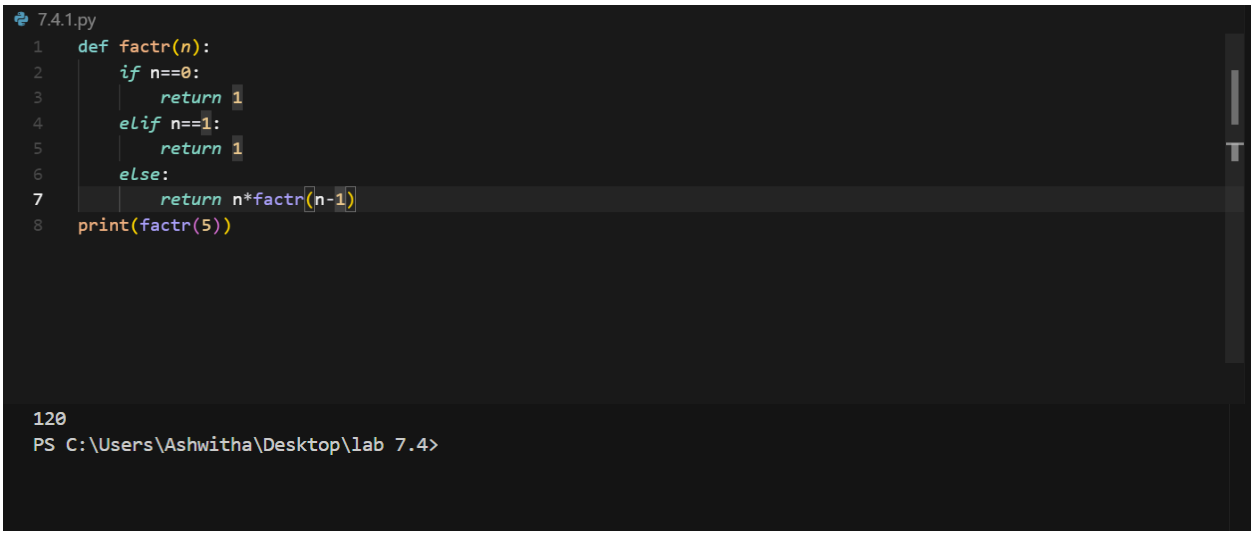
•Copilot or Cursor AI correctly identifies missing base condition or incorrect recursive call and suggests a functional factorial implementation.

CURSOR AI



VS CODE

```
7.4.1.py > ...
1  def factr(n):
2      if n==0:
3          return 1
4      elif n==1:
5          return 1
6      else:
7          return n*factr(n-1)
8  print(factr(5))
```

```
120
PS C:\Users\Ashwitha\Desktop\lab 7.4>
```

Task Description #2:

•Provide a list sorting function that fails due to a type error (e.g., sorting list with mixed integers and strings). Prompt AI to detect the issue and fix the code for consistent sorting.

```
def sort_list(data):
    return sorted(data)

items = [3, "apple", 1, "banana", 2]
print(sort_list(items))
```

Expected Outcome #2:

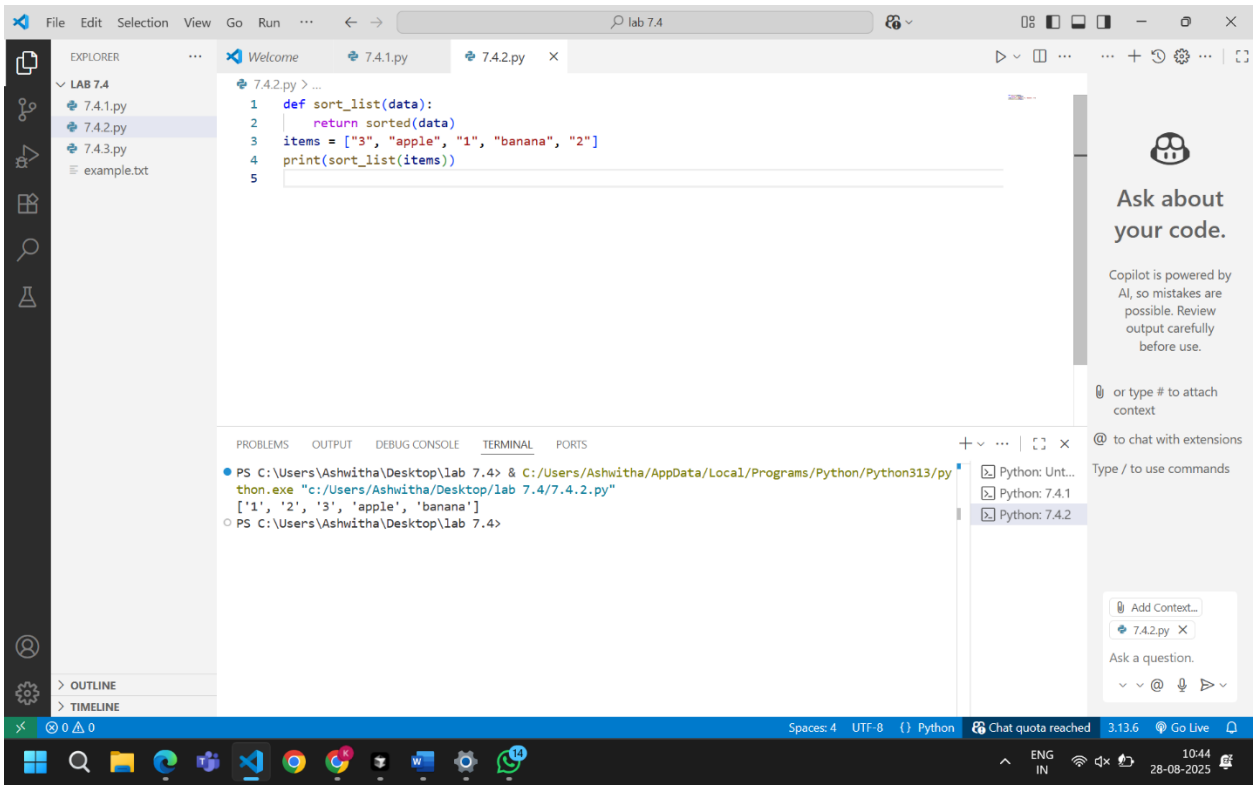
•AI detects the type inconsistency and either filters or converts list elements, ensuring successful sorting without a crash.

CURSOR AI

```
7.4.2.py
1  def sort_list(data):
2      return sorted(data)
3  items = ["3", "apple", "1", "banana", "2"]
4  print(sort_list(items))
5

['1', '2', '3', 'apple', 'banana']
PS C:\Users\Ashwitha\Desktop\lab 7.4>
```

VS CODE



Task Description #3:

• Write a Python snippet for file handling that opens a file but forgets to close it. Ask Copilot or Cursor AI to improve it using the best practice (e.g., with open() block).

Code1

```
with open("example.txt", "w") as f:
    f.write("Hello, world!")
```

Code2

```
f1 = open("data1.txt", "w")
f2 = open("data2.txt", "w")

f1.write("First file content\n")
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print("Files written successfully")
```

Code3

```
data = open("input.txt", "r").readlines()
output = open("output.txt", "w")

for line in data:
    output.write(line.upper())

print("Processing done")
```

Code4:

```
f = open("numbers.txt", "r")
nums = f.readlines()

squares = []
for n in nums:
    n = n.strip()
    if n.isdigit():
        squares.append(int(n) * int(n))

f2 = open("squares.txt", "w")
for sq in squares:
    f2.write(str(sq) + "\n")

print("Squares written")
```

Expected Outcome #3:

VS CODE

• AI refactors the code to use a context manager, preventing resource leakage and runtime warnings.

LAB 7.4 > 7.4.3.0.py > ...

```
with open("C:\\Users\\ASHWITHA\\OneDrive\\Documents\\Desktop\\AIAC\\LAB 7.4\\output.txt", "w") as f:
    f.write("Hello, world.....")
```

```
f1 = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\file.txt", "w")
f2 = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\file.txt", "w")
f1.write("First file content\n")
f2.write("Second file content\n")
f1.close()
f2.close()
print("Files written successfully.")
```

```python
data = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\file.txt","r").readlines()
output = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\file.txt","w")

for line in data:
    output.write(line.upper())

print("Processing done")
```

LAB 7.4 > 7.4.3..4.py > ...

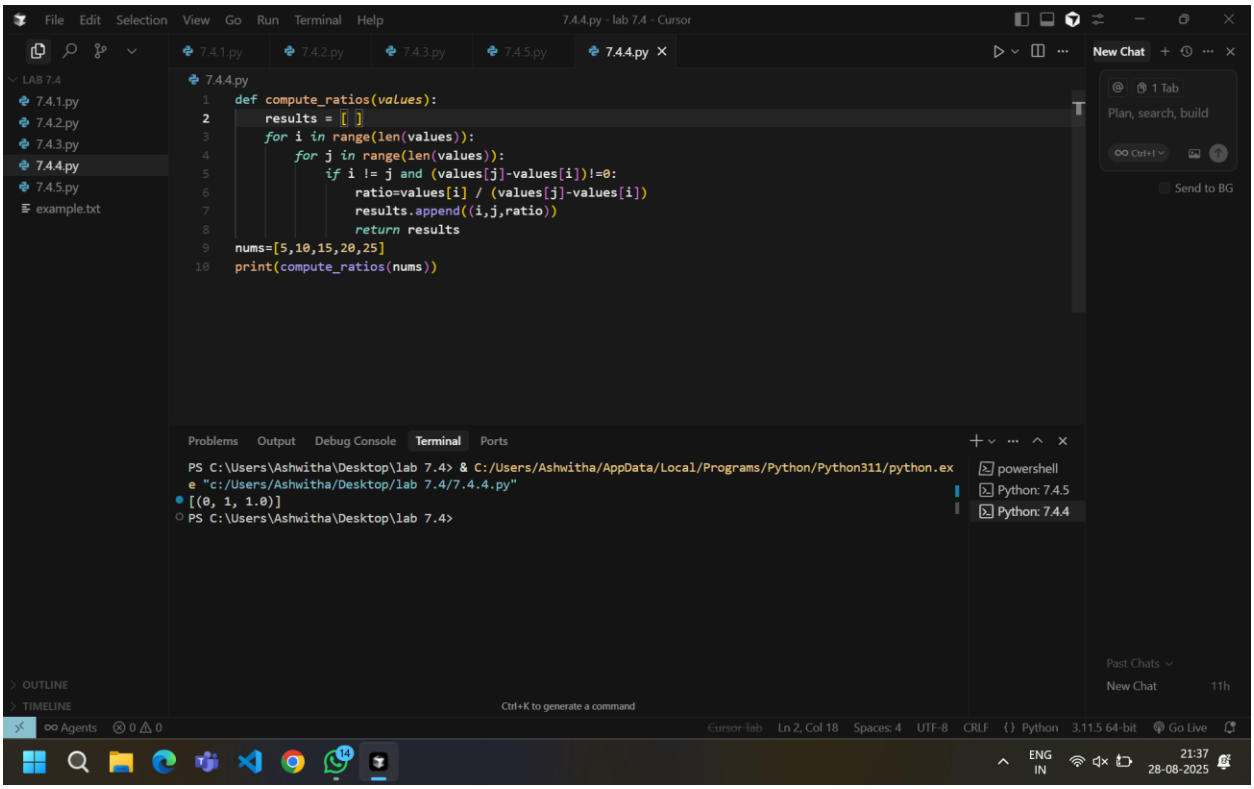
```python
f = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\file.txt", "r")
nums = f.readlines()
f.close()

squares = []
for num in nums:
    n = num.strip()
    if n.isdigit():
        squares.append(int(n) * int(n))

f2 = open(r"C:\Users\ASHWITHA\OneDrive\Documents\Desktop\AIAC\LAB 7.4\squares.txt", "w")
for square in squares:
    f2.write(str(square) + "\n")
f2.close()
print("Squares written")
```

Task Description #4:
• Provide a piece of code with a ZeroDivisionError inside a loop. Ask AI to add error handling using try-except and continue execution safely.

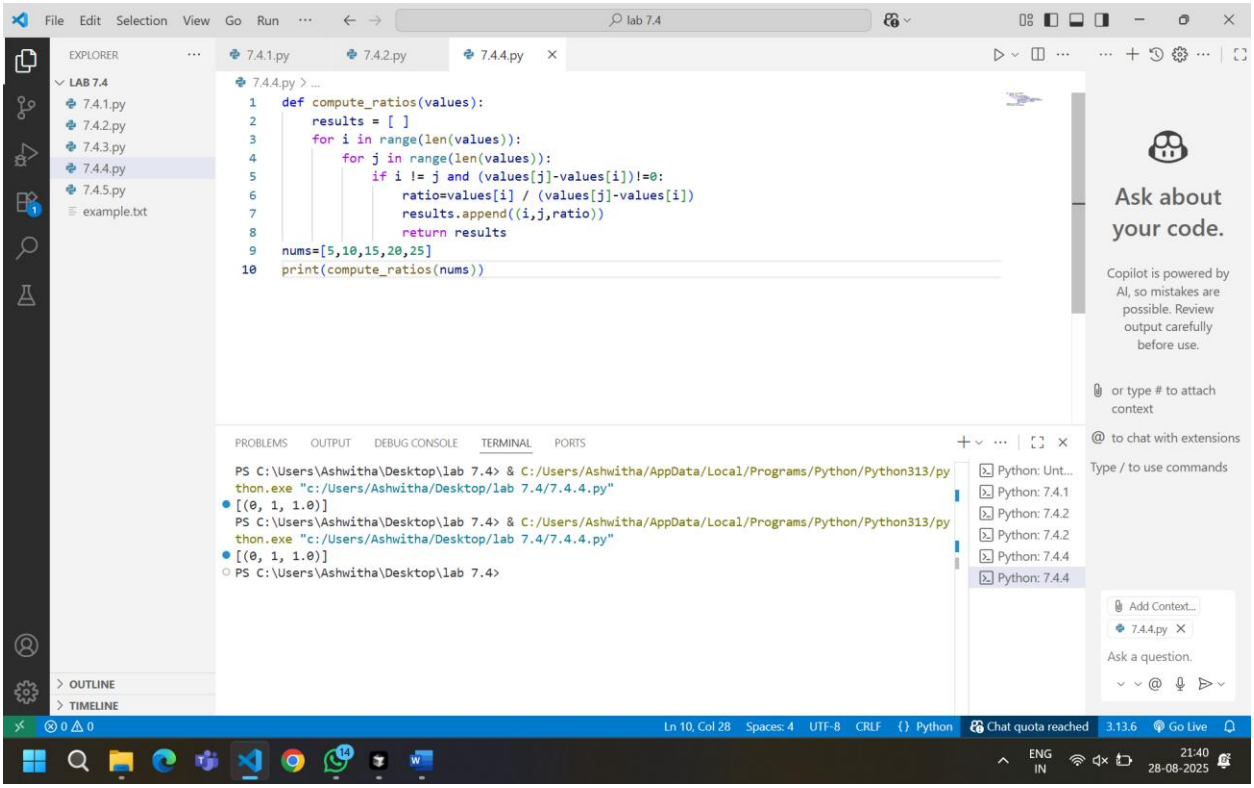
```python
def compute_ratios(values):
    results = []
    for i in range(len(values)):
        for j in range(i, len(values)):
            ratio = values[i] / (values[j] - values[i])
            results.append((i, j, ratio))
    return results

nums = [5, 10, 15, 20, 25]
print(compute_ratios(nums))
```

Expected Outcome #4:
• Copilot adds a try-except block around the risky operation, preventing crashes and printing a meaningful error message.

CURSOR AI



VS CODE



Task Description #5:

• Include a buggy class definition with incorrect __init__ parameters or attribute references. Ask AI to analyze and correct the constructor and attribute usage.

```
class StudentRecord:
    def __init__(self, name, id, courses=[]):
        self.studentName = names
        self.student_id = id
        self.courses = courseList

    def add_course(self, course):
        self.courses.append(course)

    def get_summary(self):
        return f"Student: {self.studentName}, ID: {self.student_id}, Courses: {', '.join(self.courses)}"

class Department:
    def __init__(self, deptName, students=None):
        self.dept_name = deptName
        self.students = students

    def enroll_student(self, student):
        self.students.append(student)

    def department_summary(self):
        return f"Department: {self.dept_name}, Total Students: {len(self.student)}"


s1 = StudentRecord("Alice", 101, ["Math", "Science"])
d1 = Department("Computer Science")
d1.enroll_student(s1)
print(s1.get_summary())
print(d1.department_summary())
```

Expected Outcome #5:

• Copilot identifies mismatched parameters or missing self references and rewrites the class with accurate initialization and usage

CURSOR AI

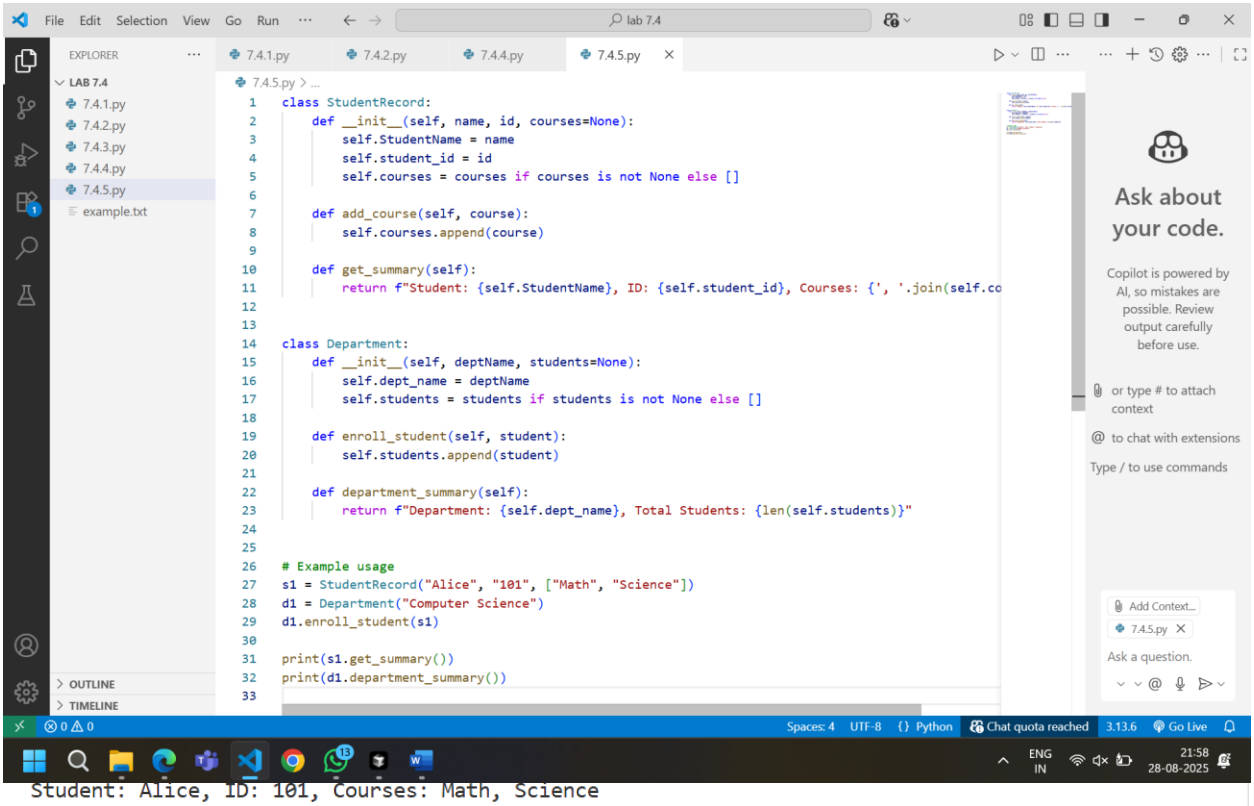
7.4.5.py

```python
class StudentRecord:
    def __init__(self, name, id, courses=None):
        self.StudentName = name
        self.student_id = id
        self.courses = courses if courses is not None else []

    def add_course(self, course):
        self.courses.append(course)

    def get_summary(self):
        return f"Student: {self.StudentName}, ID: {self.student_id}, Courses: {', '.join(self.courses)}"


class Department:
    def __init__(self, deptName, students=None):
        self.dept_name = deptName
        self.students = students if students is not None else []

```

```
    def enroll_student(self, student):
        self.students.append(student)

    def department_summary(self):
        return f"Department: {self.dept_name}, Total Students: {len(self.students)}"


# Example usage
s1 = StudentRecord("Alice", "101", ["Math", "Science"])
d1 = Department("Computer Science")
d1.enroll_student(s1)

print(s1.get_summary())
print(d1.department_summary())
```

```
Student: Alice, ID: 101, Courses: Math, Science
Department: Computer Science, Total Students: 1
PS C:\Users\Ashwitha\Desktop\lab 7.4>
```

VS CODE:




```
Student: Alice, ID: 101, Courses: Math, Science
Student: Alice, ID: 101, Courses: Math, Science
Department: Computer Science, Total Students: 1
PS C:\Users\Ashwitha\Desktop\lab 7.4>
```